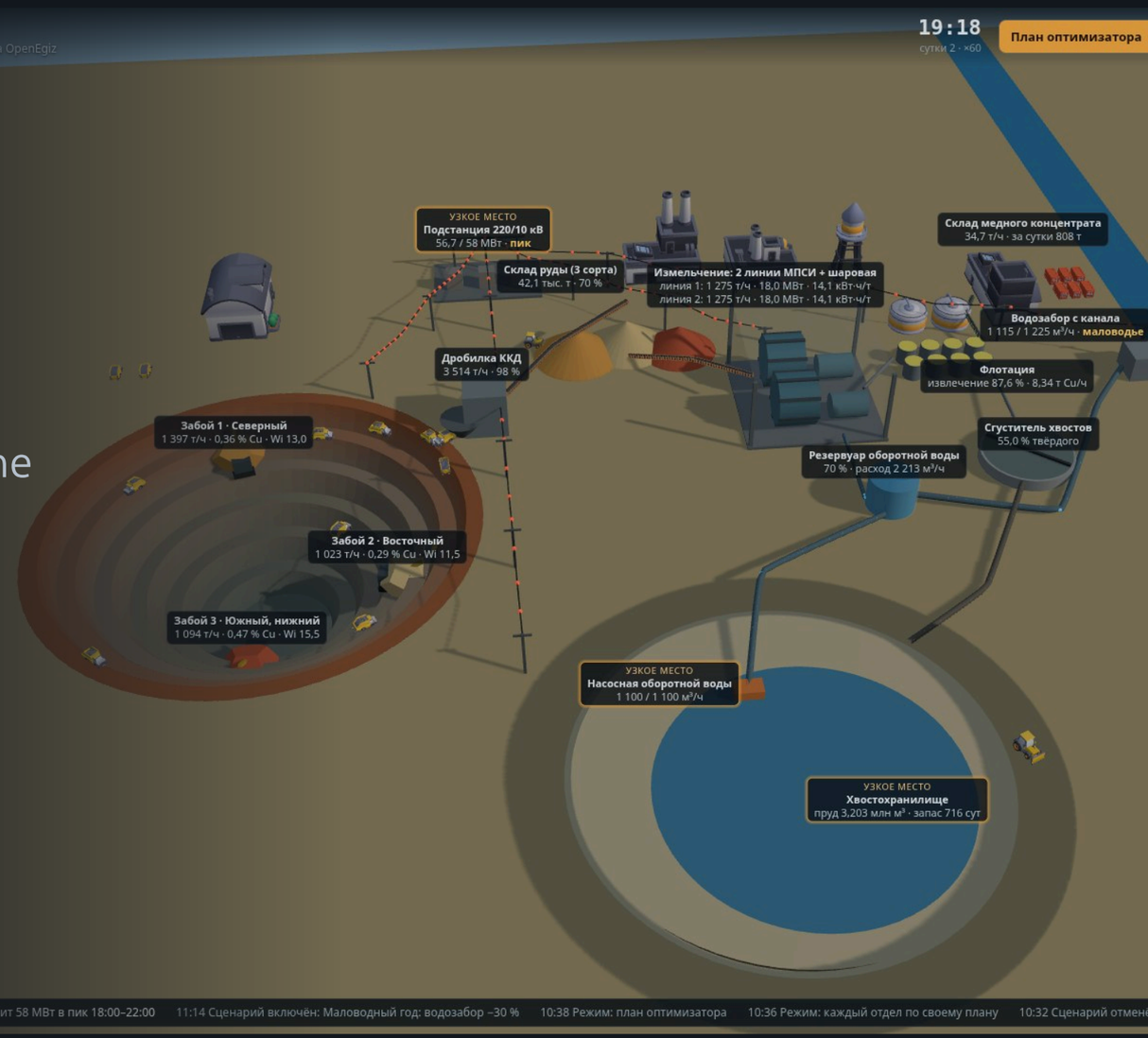
DIGITAL MINE ON OPENEGIZ · 2026

Mine Under Limits

A digital twin of a copper mine and concentrator that helps the dispatcher re-plan the day when water or power is cut

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WHY

Water and power limits hit the whole complex at once

Water

Central and East Kazakhstan are short of water, and a concentrator is one of its largest users: ~0.5 m^3 of fresh water per tonne of ore.

Power

The national grid runs at its limit. In the evening peak the capacity limit drops, and exceeding it trips a mill by automatic load shedding.

Separate departments

The pit, the plant and the tailings each plan their own area. When a limit arrives, the plan is cut by eye and in proportion.

The limits will only tighten, while decisions today are made department by department.

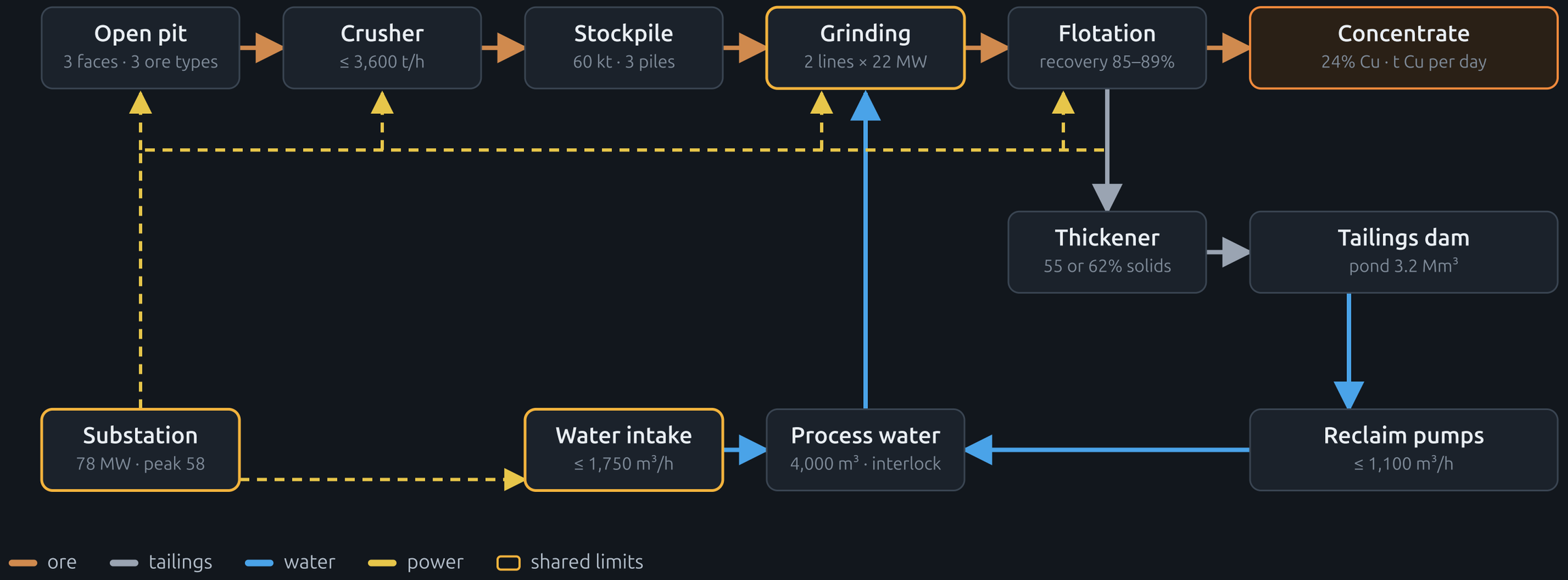
CONCEPT

One twin for the whole complex — from pit face to tailings dam

Object	A full-cycle porphyry copper complex: open pit (3 faces), crusher, conveyor, stockpile, concentrator (2 grinding lines, flotation), thickener, tailings dam with water reclaim, water intake, substation
User	The head of the production dispatch office, who draws up the daily and shift plan
Decision	How to rebuild the mining and processing plan under a limit: how much ore to dig and where, how hard to load the mills, how much water to reclaim from the pond — and what each choice costs in copper
Real or fictional	Fictional, at the scale of an Aktogay sulphide concentrator; grade, recovery and water use checked against open KAZ Minerals data

MINE MODEL

The complex as one system: ore, water and power are linked



23 twins in Eclipse Ditto, namespace kz.limit. Links live in the twins' own attributes: _parents (hierarchy), ore_to, water_to, water_from, power_from (flows).

BEHAVIOUR

States are linked by physics, not by random numbers

Hardness → energy

```
E = 1.35 · 10 · Wi · (1/√150 − 1/√100,000)
```

Bond’s law. A mill is capped by its 22 MW drive: hard ore cuts throughput by itself.

Feed → water

```
0.87 m³ of water per t of ore
```

Water leaves with the tailings. Process tank below 10% — the interlock cuts plant feed. High-density thickening saves 0.2 m³/t for 0.4 kWh/t.

Load → shedding

```
Σ loads > limit
```

Shovels, crushing, mills, flotation, pumps. Above the limit, load shedding trips the most loaded line for 40 minutes.

Stockpile decouples

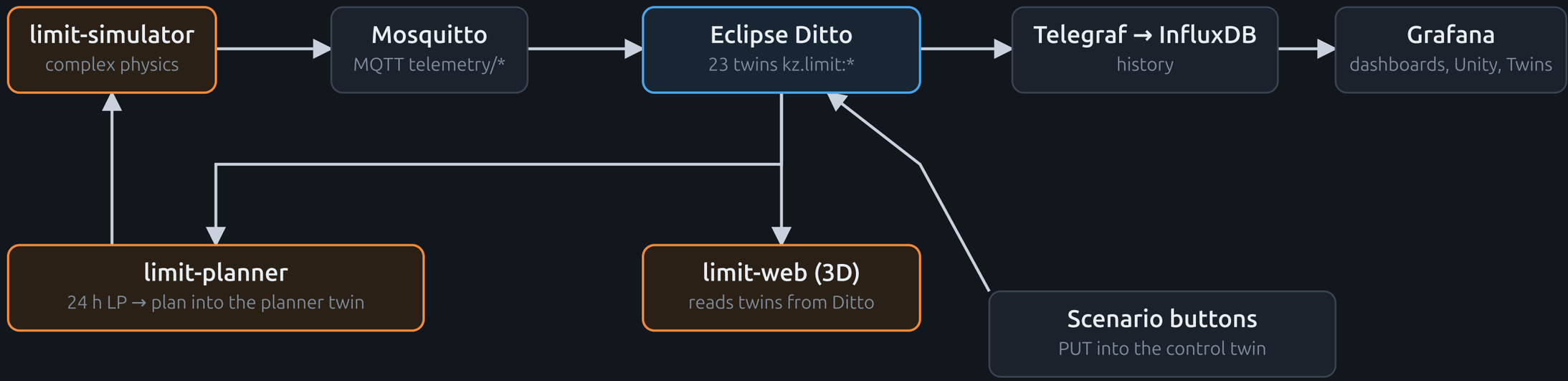
```
pit ≠ plant
```

In the peak the plant cuts feed while the pit keeps digging into the stockpile. The tailings pond is a water buffer for days and weeks.

The simulator steps 2 model minutes every 2 seconds (×60). Every parameter is in one file, team/engine/params.py, shared by simulator and optimizer.

PLATFORM

Built on OpenEgiz, not next to it



- Twins are written only MQTT → Ditto, like a real sensor. The optimizer publishes its plan the same way.
- 3D, Grafana and the Twins app read state from the platform. A what-if scenario is a write into the control twin.
- Everything starts with one make up and is checked by make smoke. No external services or keys.

MODULE

Daily plan optimizer: a 24-hour linear program

- For each of the next 24 hours: digging per face, mill feed per ore type from the stockpile, thickener mode, fresh and reclaimed water
- Goal — maximum copper in concentrate over the day
- Limits: hourly substation power (incl. peak), intake, pumps, pond, mills, crusher, stockpile, mine plan ±20% per ore type
- SciPy / HiGHS, ~30 ms. Re-solved every 15 model minutes, on the hour and at once on any scenario change

Plan	setpoints go to the simulator — the module runs the mine
By departments	same problem with department rules: the plan's gain in t Cu
No limits	what the limits cost in tonnes of copper
Shadow prices	+1 MW in peak, +100 m^3/h water, +100 t/h mills — in tonnes of copper

Three solves per re-plan. The dispatcher's advice is built from the solution by templates — no language model, no external API.

VISUALIZATION

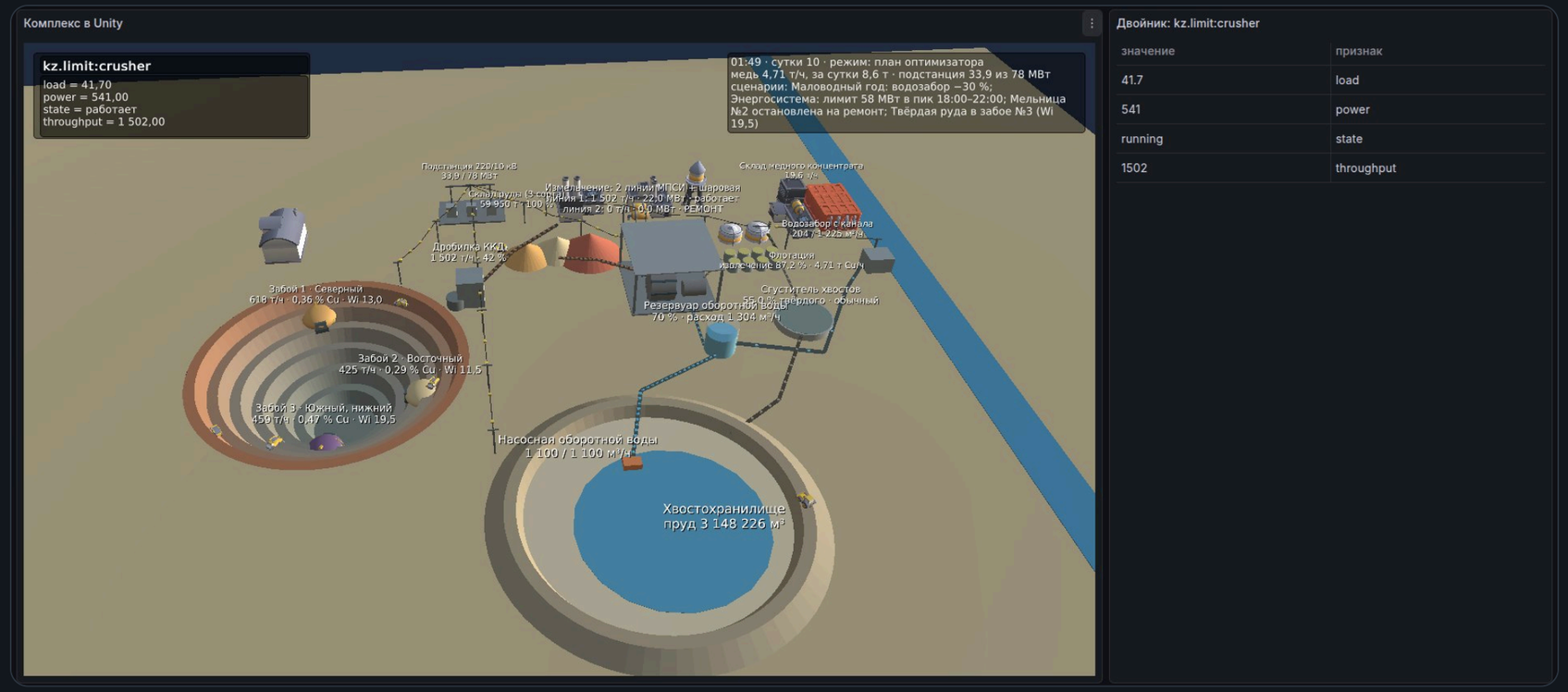
3D view of the complex: http://localhost:8100



- Ore, water and power flows are animated in proportion to the twins' values
- Bottlenecks found by the optimizer get an amber frame; a line under repair is grey, a tripped one red
- Stockpile cones per ore type, tank level, pond area, time of day — from Ditto every 2 seconds
- Haul trucks (loaded up, empty down, spare ones at the workshop), concentrate shed, site buildings — ready-made Kenney models (CC0)

VISUALIZATION IN GRAFANA

The complex in the Unity panel



- Grafana → «Рудник под лимитом — Unity 3D»: the same scene in the OpenEgiz Unity panel, a Unity 6 WebGL build
- Objects are named after the twins; on every dashboard refresh the panel sends each one its values from InfluxDB
- Clicking an object shows its twin card; the SelectTwin event writes it into a dashboard variable, and the table next to it lists every value
- The scene is generated in code from the same layout and Kenney models; make unity-on / make unity-off switch the dashboard

FOR THE DISPATCHER

Advice, three plans and 24-hour charts next to the scene



Dry year + evening peak, «Optimizer plan» mode. Left: resources and scenarios; right: advice and plan; bottom: event log. The interface is in Russian.

WHAT-IF

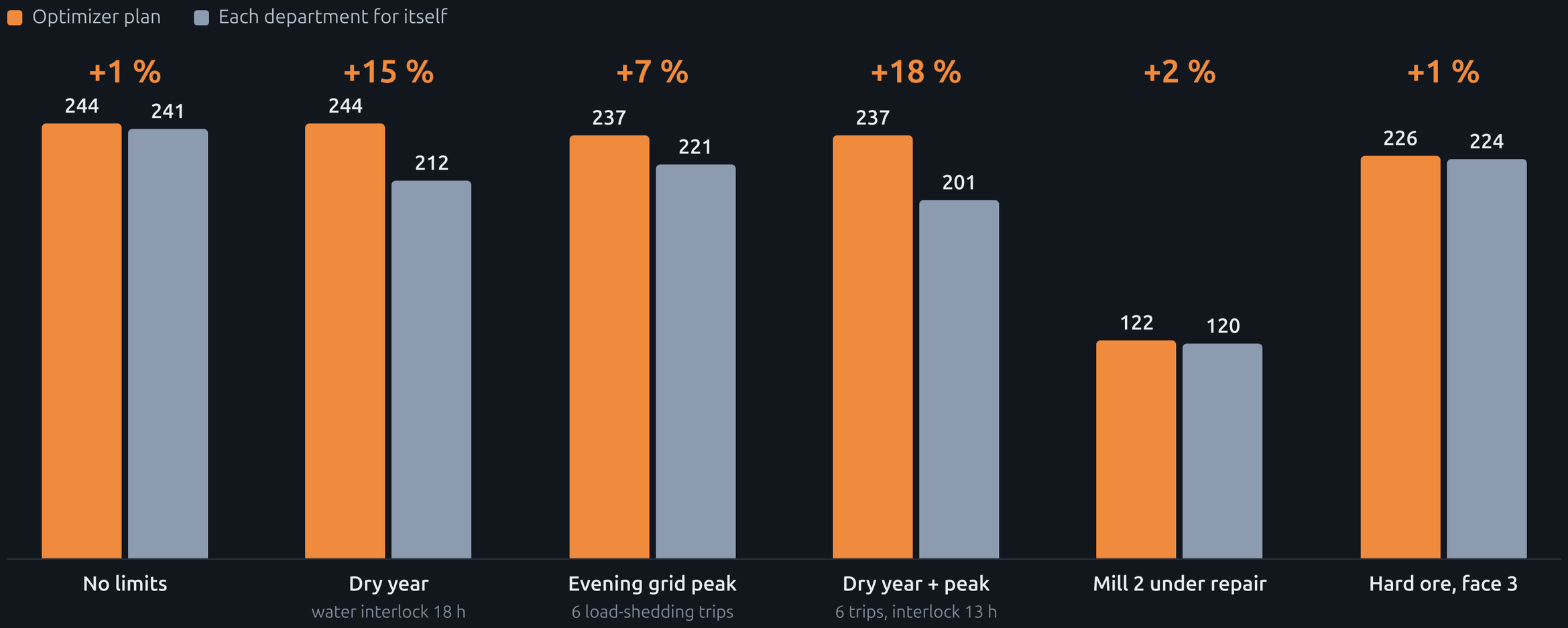
Four scenarios: switched on and undone with one button

SCENARIO	WHAT CHANGES	WHAT THE OPTIMIZER DOES
Dry year `dry_year`	intake −30%: 1,750 → 1,225 m^3/h	switches thickening to high density and keeps the feed
Evening peak `grid_peak`	limit 78 → 58 MW, 18:00–22:00	cuts mill feed ahead of time, the pit digs into the stockpile
Mill 2 under repair `mill2_repair`	one grinding line of two	re-plans the day and the blend for one line
Hard ore, face 3 `hard_ore`	Wi 15.5 → 19.5 kWh/t	lowers the hard ore share within the mine plan

Scenarios combine. A button writes into the kz.limit:control twin; the same works with a curl PUT to Ditto. The «Each department for itself» switch shows the complex running without the plan.

RESULT ON THE MODEL

Copper per day, t: the gain is where the limit is



One model day from 16:30, the same simulator; make evaluate recomputes it (~10 s). Without limits the plan nearly matches the departments: the optimizer may not leave the mine plan by more than ±20%.

SIDE BY SIDE

Dry year + peak: plan vs «each department for itself»



Optimizer plan: feed cut in advance, no trips

8.33
t Cu/h at 19:30

56.9
MW, limit 58

0
load-shedding trips

237
t Cu per day



Each department: load shedding tripped grinding line 1

5.01
t Cu/h at 19:54

1 of 2
lines running

6
trips per day

201
t Cu per day

DATA AND ASSUMPTIONS

Synthetic data, scale checked against open sources

PARAMETER	IN THE MODEL	SOURCE
Throughput	2 × ~1,560 t/h ≈ 25 Mt/y	one Aktogay sulphide plant (KAZ Minerals)
Cu grade	0.29 / 0.36 / 0.47 %	Aktogay: 0.32% in resources, 0.50% milled in 2023
Recovery	85–89 %	Aktogay sulphide plant: 88%
Bond work index	11.5 / 13.0 / 15.5 (19.5) kWh/t	911 Metallurgist table: copper ore 12.7–14
Fresh water	~0.52 m^3/t	between Bozshakol (0.43) and Aktogay (0.60), 2024
Water in tailings	0.82 / 0.61 m^3/t	computed from 55 / 62% solids
Limits	78 (peak 58) MW · 1,750 m^3/h	Team assumptions, fitted to the unit consumptions

Full table with links: SUBMISSION.md, «Data and Assumptions». No confidential or personal data.

HONESTLY

Limitations and next steps

What is simplified

- Hourly balances: no truck cycles, drill and blast, flotation kinetics
- Recovery per ore type is constant; the goal is tonnes of copper, no prices or penalties
- Plan horizon 24 hours; evaporation seasonality simplified

What is next

- Money in the goal: copper price, flocculant, capacity tariff
- A weekly horizon and the pond as a seasonal water buffer
- Real plant data through the same MQTT → Ditto path

TEAM

team-101 and how to run it

khusrav11	Complex model: simulator and twins
BlackIc32007	Daily plan optimizer module
bekbolatnurmas-coder	3D visualization: Three.js and the Grafana Unity panel
Dauka-dr	Integration, Grafana, docs, presentation

```
sudo apt-get update && sudo apt-get install -y make
make up
make smoke
```

3D	http://localhost:8100
Grafana	http://localhost:3000 → Рудник под лимитом dashboard